CLASS: SY-B ENTC

PRN: 25UET070

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SLA-4:

Problem Statement 15: Autonomous Multi-Agent Strategy Planning

Developing effective plans in complex environments requires agents to evaluate alternative actions and respond to unexpected changes. This project aims to design an autonomous multi-agent strategy planning system that analyzes the current environment, generates alternative plans, evaluates their outcomes, and selects suitable actions to achieve specified goals.

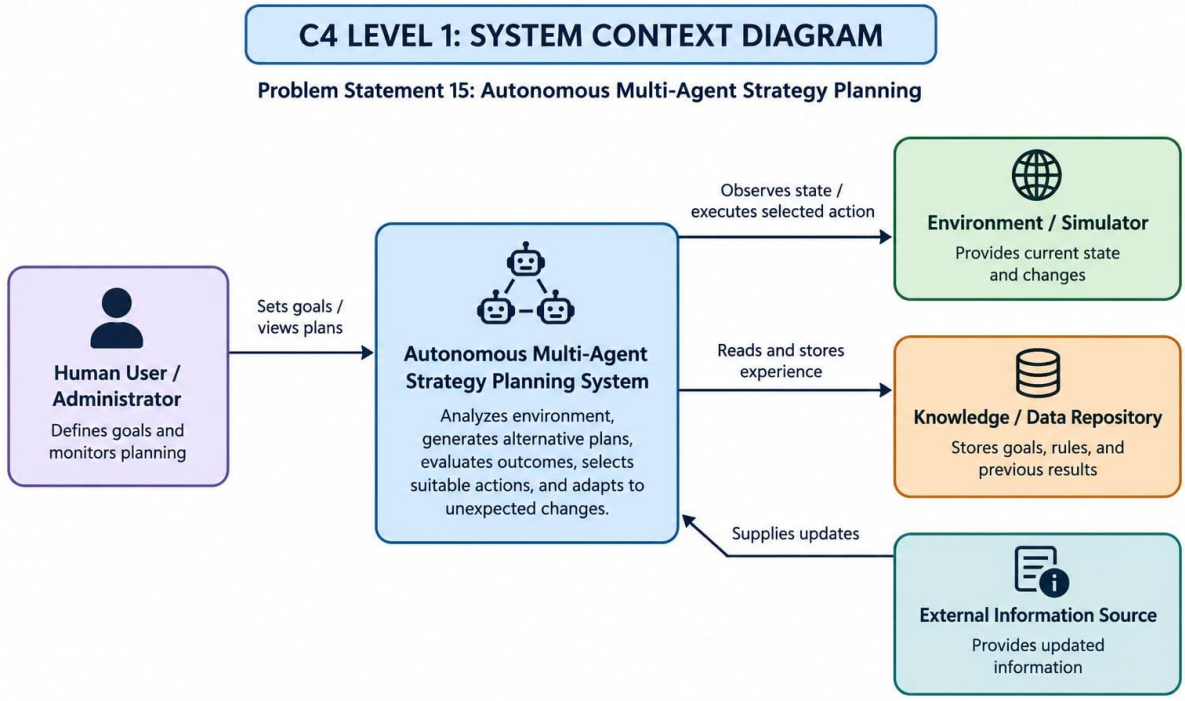
INTRODUCTION:

The Autonomous Multi-Agent Strategy Planning System is designed to solve complex problems by allowing multiple intelligent agents to plan and make decisions automatically. In changing environments, agents need to understand the current situation, consider different possible actions, and select the most suitable plan.

This system helps agents analyze environmental conditions, generate alternative strategies, predict possible outcomes, and choose actions according to the desired goals. When unexpected changes occur, the agents can update their plans and adapt their decisions.

The system uses a structured architecture to organize its different modules and improve communication between agents. The main objective is to achieve goals efficiently, reduce planning errors, and improve decision-making in complex environments.

C4 LEVEL 1 :

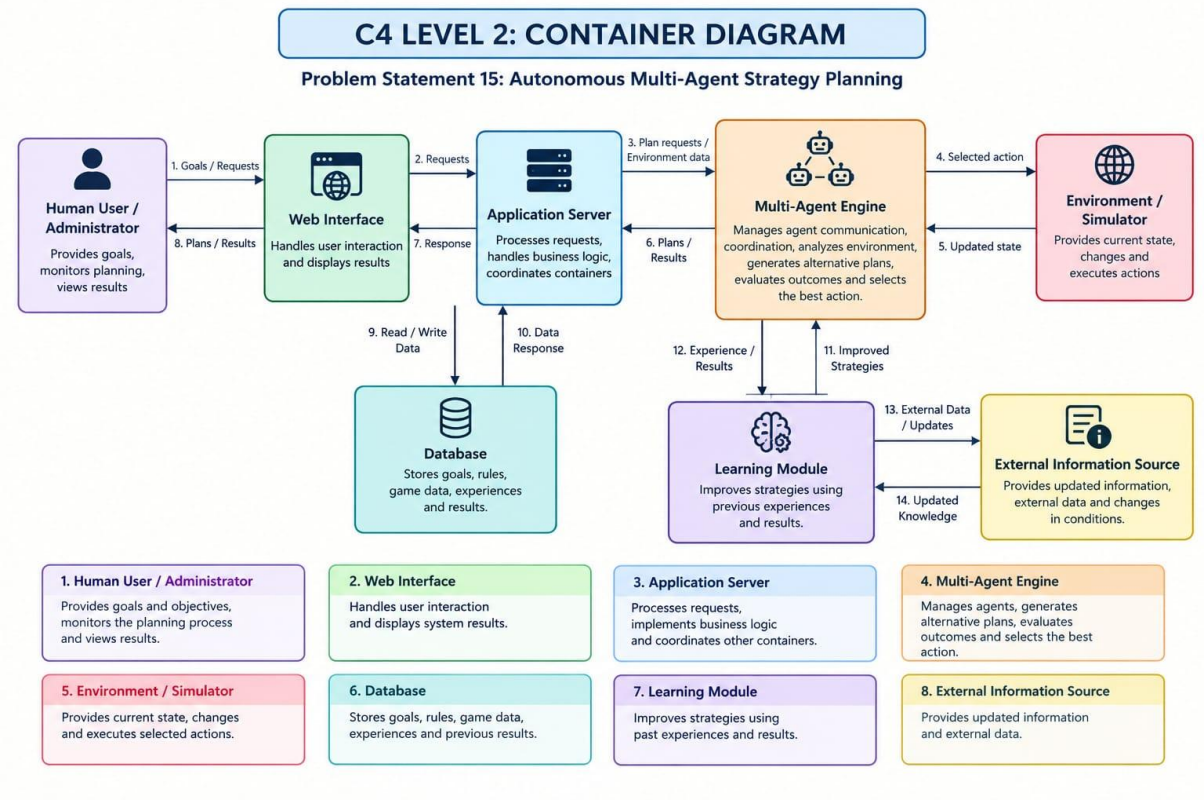


The diagram represents the overall structure of the Autonomous Multi-Agent Strategy Planning System and shows how it interacts with external entities. It provides a high-level view of the system without showing its internal technical details. The main purpose of this system is to analyze the current environment, generate alternative plans, evaluate their possible outcomes, and select suitable actions to achieve specified goals.

The diagram consists of five main elements: Human User/Administrator, Autonomous Multi-Agent Strategy Planning System, Environment/Simulator, Knowledge/Data Repository, and External Information Source.

The Human User or Administrator interacts directly with the system. The user defines the goals and objectives that the system must achieve. The administrator can also monitor the planning process and view the selected plans. The goals provided by the user act as input to the system, helping it understand what needs to be accomplished. Based on these goals, the system analyzes the available information and prepares suitable strategies.

C4 LEVEL 2:



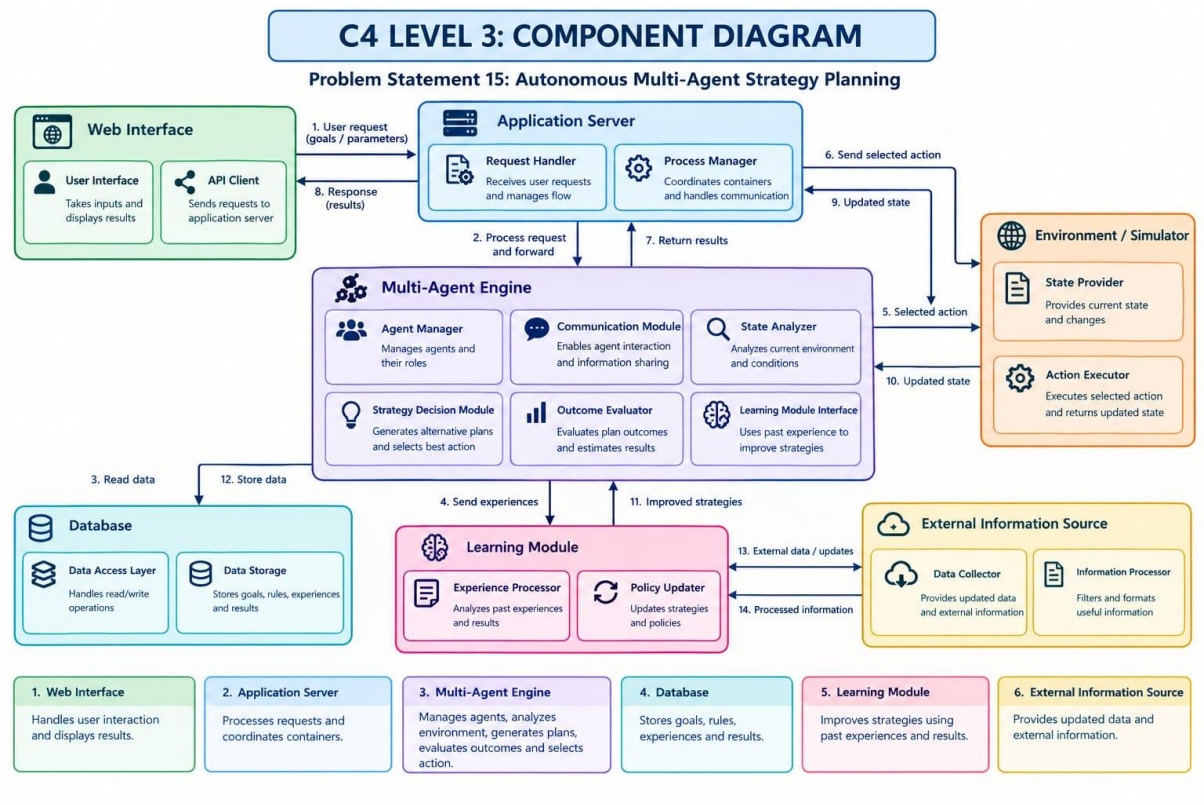
The diagram represents the internal structure of the Autonomous Multi-Agent Strategy Planning System. It explains how the system is divided into different containers and how these containers communicate with each other to achieve the specified goals. Each container performs a specific function, such as handling user requests, processing information, generating plans, storing data, or improving strategies. This diagram provides more technical details in the diagram.

The diagram consists of eight main containers: Human User/Administrator, Web Interface, Application Server, Multi-Agent Engine, Environment/Simulator, Database, Learning Module, and External Information Source.

The Multi-Agent Engine is the main part of the system. It analyzes the environment, generates alternative plans, evaluates possible outcomes, and selects suitable actions. The Environment/Simulator provides the current state and updates the system after an action is executed.

The Database stores goals, rules, previous experiences, and results. The Learning Module uses previous experiences to improve planning strategies. The External Information Source provides updated information that helps the system adapt to changing conditions.

C4 LEVEL 3:



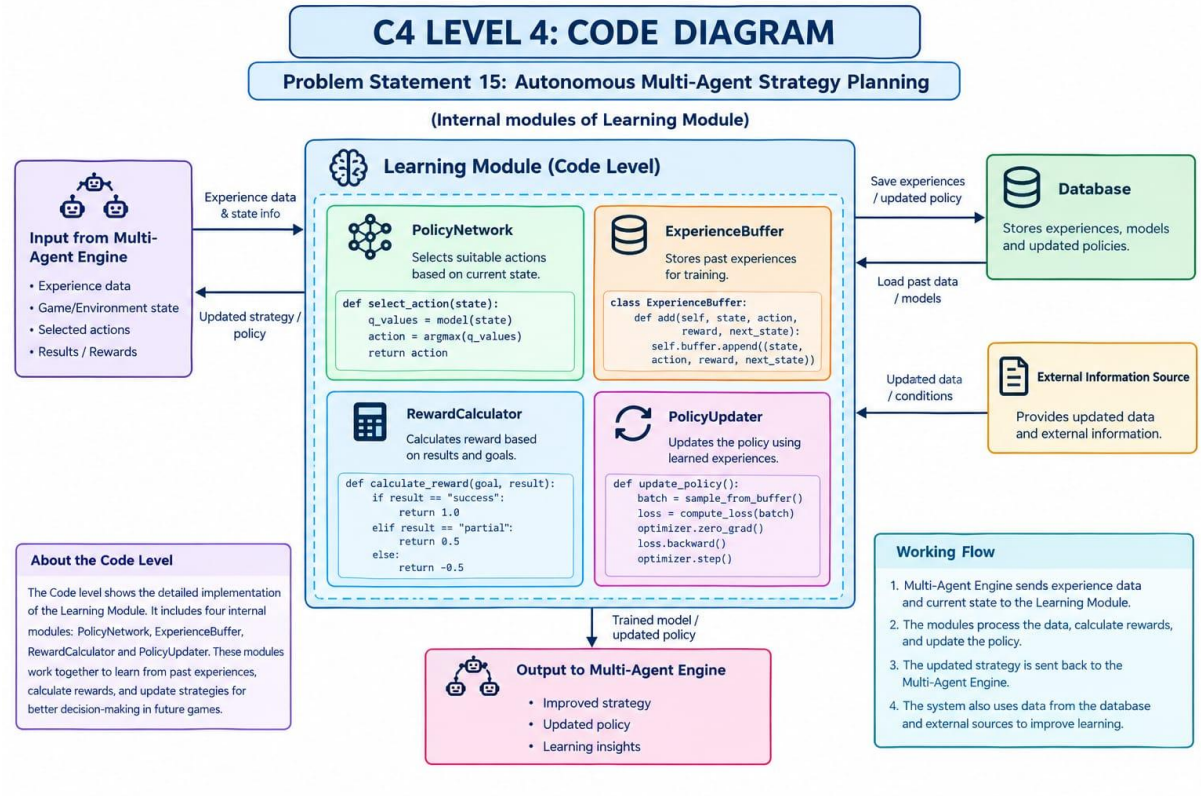
The C4 Level 3 diagram explains the internal components of the Autonomous Multi-Agent Strategy Planning System. It focuses mainly on the Multi-Agent Engine and shows how its components work together to analyze the environment, generate plans, evaluate outcomes, and select suitable actions. It also shows the interaction of the engine with the Application Server, Database, Environment/Simulator, Learning Module, and External Information Source.

Main Components

Web Interface: It accepts user inputs and displays the results. The User Interface handles interaction with the user, while the API Client sends requests to the Application Server.

Application Server: It receives user requests and manages the flow of information. The Request Handler processes requests, while the Process Manager coordinates communication between different components.

C4 LEVEL 4:



The C4 Level 4 diagram represents the detailed implementation of the Autonomous Multi-Agent Strategy Planning System. It focuses on the internal modules of the Learning Module and explains how they process experience data, calculate rewards, and update strategies. It shows how different code-level components work together to improve the decision-making ability of autonomous agents.

Input from Multi-Agent Engine

The Multi-Agent Engine provides experience data and the current state of the environment to the Learning Module. This information includes selected actions, previous results, and rewards. The Learning Module uses these inputs to understand how well the agents are performing and identify areas for improvement.